

SHRINKING THE DIGITAL HEAP:

TRIAGING DIGITAL CARRIERS AT CAMBRIDGE UNIVERSITY LIBRARY

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Background

Over 1550 digital carriers have been identified in Cambridge University Library's (CUL's) archival collections, which are managed by a team of archivists.

The Technical Analyst works to transfer the contents of digital carriers to the digital preservation repository.

Carriers often have unclear or no labelling, making it challenging to determine processing priorities or to identify carriers that do not need to be transferred.

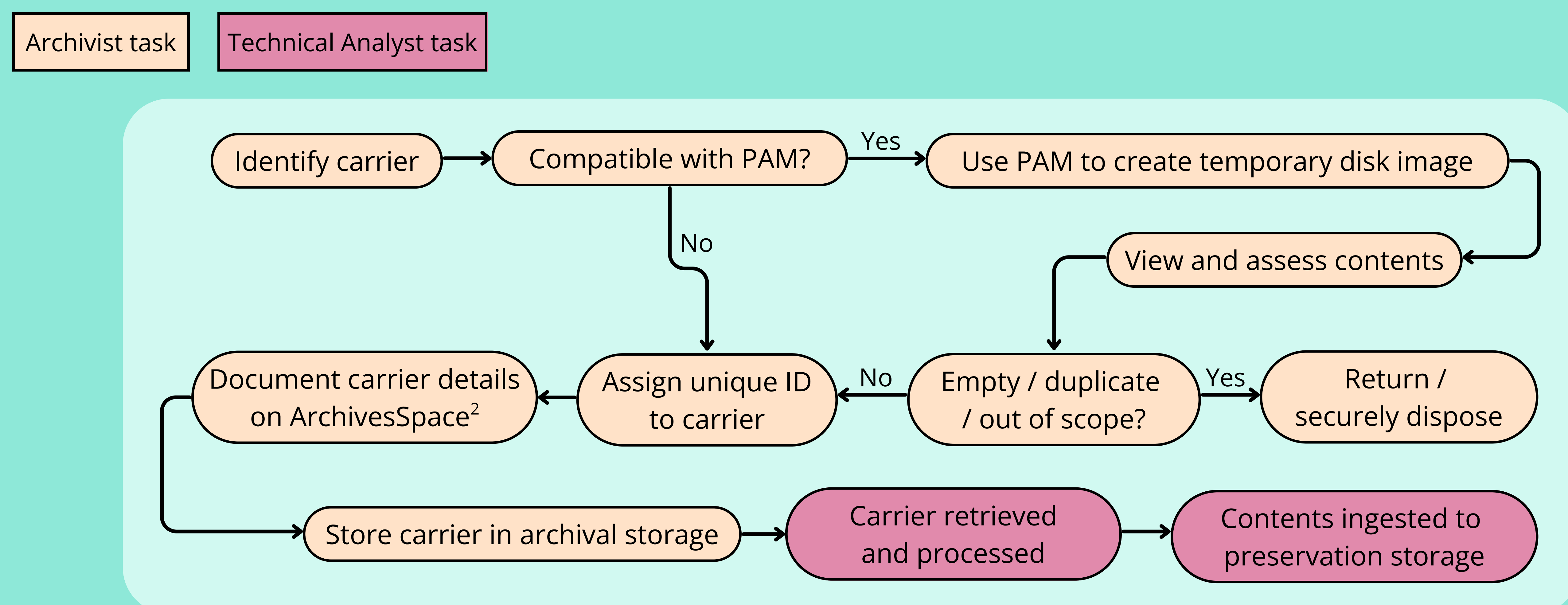
After processing, carriers were sometimes found to be blank, duplicates, or out of scope for permanent preservation.

It is estimated that it would take over 4 years for the Technical Analyst to process all of CUL's archival digital carriers. This does not include carriers held by other collecting areas of the library.

Aim

Develop a workflow for archivists to assess and triage digital carriers before sending them for processing.

Workflow



Quantitative Outcomes

Digital carriers assessed using PAM since early 2025: **over 230**.

Carriers found to be empty, duplicates or out of scope: **146**.

Archivist working time taken to pre-appraise carriers: **approximately 7 days**.

Technical Analyst processing time saved: **approximately 5 months**.

Approach: Pre-Appraisal Machine (PAM)

- Decommissioned desktop PC obtained from IT team
- Installed with BitCurator¹ environment, including disk imaging tools and software write blocker



- USB 3.5-inch floppy disk drive added
- Not connected to library network or internet

- Compatible with 3.5-inch floppy disks, optical discs, USB drives and external hard drives smaller than 200GB
- Set up in archivists' working area
- User guidance produced by Digital Archivist and Technical Analyst
- 3 training sessions for archivists on using PAM were held

Qualitative Outcomes

Before implementing this workflow, the archivists had little engagement with practical digital preservation work.

Now they are familiar with different types of digital carriers, the process of creating disk images, and the file formats and structures they might expect to see on legacy carriers.

References

- <https://github.com/BitCurator/bitcurator-distro/wiki/Releases>
- <https://archivesearch.lib.cam.ac.uk>